

ndimensions labs: SmolVLA 1-Day Challenge

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1 Dataset

I selected LIBERO-Spatial Task 0 (*“pick up the black bowl next to the cookie box and place it on the plate”*) as the training task. LIBERO-Spatial is used in the original SmolVLA evaluation, providing a reference benchmark, and its multiple tasks ensure easier OOD testing. The dataset contains 43 episodes (5,192 frames, mean episode length 120.7 frames), split into 38 for training and 5 for validation. The observation space consists of a third-person RGB image, a wrist RGB image (both 256x256), and an 8-dimensional proprioceptive state. The action space is a 7-dimensional continuous end-effector delta command.

2 Fine-tuning

I fine-tuned `HuggingFaceVLA/smolvla-libero`, pre-trained on the full LIBERO benchmark, rather than `smolvla_base` which is trained on SO100 community data. Starting from `smolvla-libero` eliminates the cross-embodiment gap and ensures the model already understands the Franka Panda action space and LIBERO scene semantics. I applied LoRA with rank $r=16$ to the action expert’s attention layers, keeping the VLM backbone fully frozen, resulting in 588K trainable parameters out of 606M total (0.1%). Training used AdamW with peak learning rate $1e-4$, cosine decay to $2.5e-6$, 1,000 warmup steps, batch size 48, and 10,000 steps. The model is trained with a flow matching objective, minimizing MSE between predicted and target vector field directions over the action chunk.

3 Training Diagnostics

Training loss decreased smoothly from 0.142 at step 100 to 0.056 by step 10,000, plateauing around step 7,000. GPU memory stabilized at 20GB out of 24GB, with data loading rather than compute being the primary bottleneck. Validation loss was computed post-hoc on saved checkpoints, with runs showing rising validation loss after step 4,000, indicating overfitting on the small 43-episode dataset.

4 Evaluation

I evaluated the fine-tuned policy over 10 rollouts per condition in the LIBERO MuJoCo simulation. The tasks tested were:

- Task 0: pick up the black bowl next to the cookie box and place it on the plate
- Task 1: pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate
- Task 2: pick up the black bowl on the ramekin and place it on the plate

Condition	Description	Success Rate
ID	Task 0 (trained task)	30%
OOD-1	Task 1 (within pretrained dist)	70%
OOD-2	Task 2 (within pretrained dist)	20%

Table 1: Policy evaluation results.

ID performance of 30% is modest and probably below what `smolvla-libero` would achieve zero-shot, suggesting that fine-tuning on 43 episodes introduced task-specific bias without actually improving on the pretrained baseline. OOD-1 at 70% exceeded ID, with the frozen VLM retaining pretraining knowledge of seen tasks. OOD-2 was also low at 20%, perhaps due to fine-tuning narrowing the action distribution. Tasks

0 and 2 both involve bowl placement, so specializing on Task 0 may have degraded performance on Task 2 more than on the less related Task 1.

5 Ablation Study

I compared LoRA rank $r=8$ vs $r=16$ with all other hyperparameters held constant (batch size 48, 10,000 steps, learning rate $1e-4$).

LoRA Rank	Final Train Loss	ID Success Rate
$r=8$	0.056	0%
$r=16$	0.056	30%

Table 2: Ablation: effect of LoRA rank.

$r=8$ resulted in 0% ID success rate, with rollout videos showing the arm repeating a fixed action sequence throughout episodes. Rising validation loss throughout training confirms overfitting, suggesting $r=8$ and even $r=16$ are insufficient for finetuning effectively. $r=16$ still worked slightly better, achieving 30% ID success while retaining OOD-1 performance at 70%.

6 What I would do with more time

With more time, I would address the validation monitoring gap first: the val loss was only logged post-training, so overfitting in both runs went undetected until after training completed. Online validation with early stopping would have caught this and allowed for more informed rank selection. I would start with much higher ranks ($r=32$, $r=64$) to allow the action expert to adapt better, and increase the number of training episodes by incorporating semantically similar tasks alongside Task 0. With more compute, I would evaluate across all 10 LIBERO-Spatial tasks rather than 3, run more rollouts per condition to ensure fair evaluation of finetuning performance.